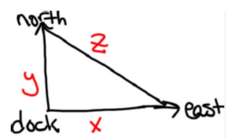
Problem 1

Problem 1

Problem 1 *Find the mistake in the solution to the given problem. Then solve the problem correctly.*

Two boats leave from the same dock, but at slightly different times. One boat is traveling east at 30mph while the other boat is traveling north at 15mph. At an instant in time when the boat traveling east is 15 miles from the dock and the boat traveling north is 10 miles away from the dock, at what rate is the distance between the boats changing?

- We want to find $\frac{dz}{dt}$ so
  $x^2+y^2=z^2$
- We know $x=15, y=10,$
  $\frac{dx}{dt}=30, \frac{dy}{dt}=15$
- Substituting $x=15$ and $y=10$:
  $15^2+10^2=z^2$ or $325=z^2$



- Take the derivative:
  $0=2z\frac{dz}{dt}$
- So $\frac{dz}{dt}=0$